PHYS512: Introduction to Machine Learning PT 2

Guest Lecturer: Maxwell A. Fine



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About me:



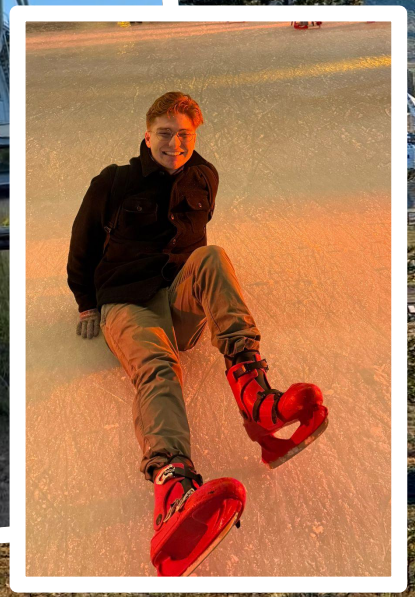




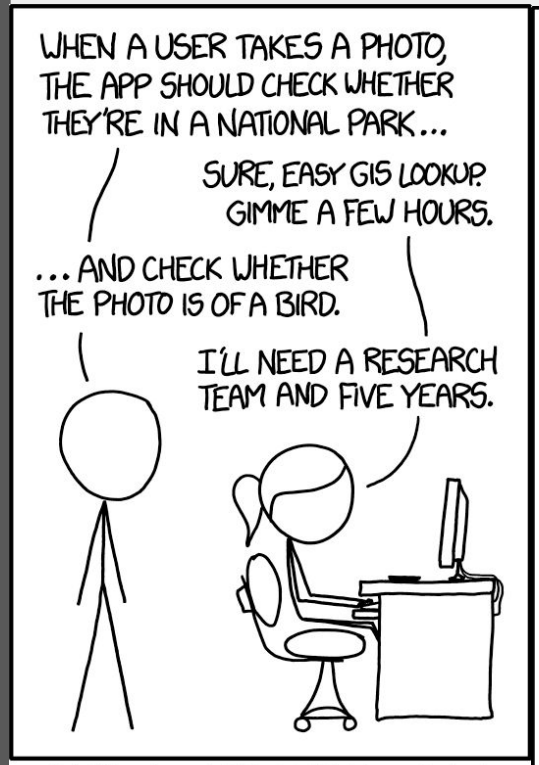
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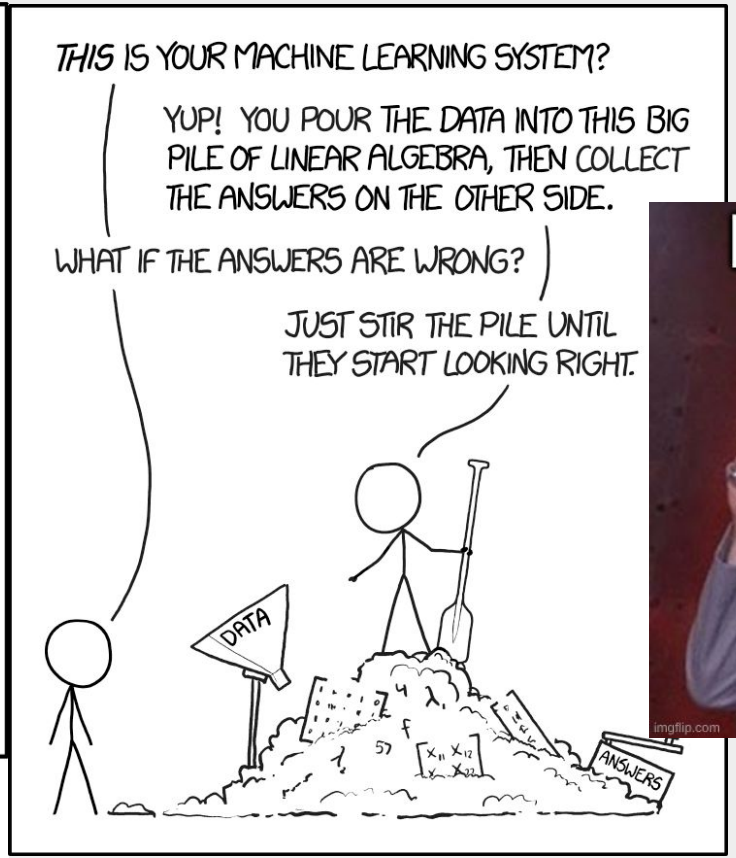


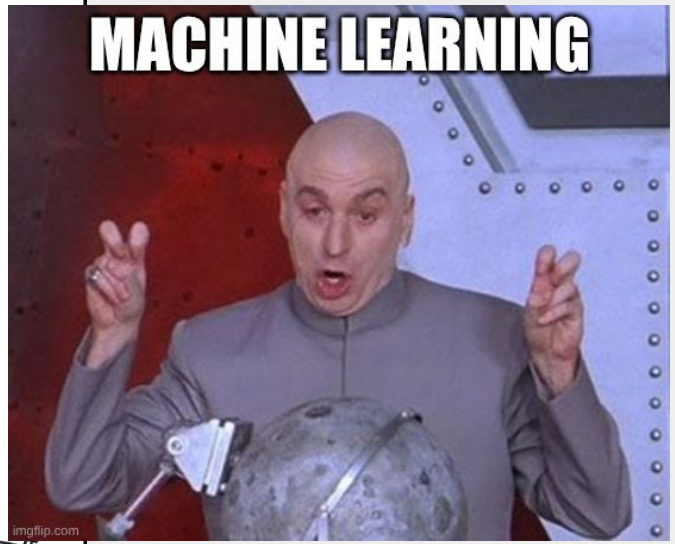






IN CS, IT CAN BE HARD TO EXPLAIN THE DIFFERENCE BETWEEN THE EASY AND THE VIRTUALLY IMPOSSIBLE.





Todays Outline:

- Recap part 1
- Machine Learning Workflow
- Training and overfitting
- Evaluating models
- Pitfalls in Machine Learning

- In class exercise: MNIST

ML in Physics: Identify the type of problem

Observations

1. Event filtering at **Large Hadron Collider**
2. Noise reduction and signal enhancement by **Event Horizon Telescope**
3. Anomaly detection in **astronomical surveys**

Modeling

1. **Emulating** complex simulations (e.g., fluid dynamics, climate models)
2. **Physics-Informed Neural Networks (PINNs)** to solve complex PDEs more efficiently than traditional numerical methods

Inference

1. Masses and spins of black holes in gravitational wave detections by **LIGO**
2. Phase transitions into new states of matter, such as **superconductors**

Prediction

1. **Weather and climate** forecasting
2. Predict the existence of and search for **exotic new particles**
3. Predict which **FRBs** are highly likely to repeat in the future

Sources: Albrecht et al., 2025, J. Phys. G: Nucl. Part. Phys. 52 030501 · Janssen et al., 2025, A&A, 698, A62 · Etsebeth et al., 2024, MNRAS, 529(1), 732-747 · Suresh R, et al., 2024, Front. Phys., 12:1322162 · George & Huerta, 2018, Physics Letters B, 778, 64-70.

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A rule of thumb

A starting point, not a strict recipe

Model-based fitting

e.g., regression, MCMC. Reach for it when:

- You can write down a physical model and its likelihood
- You want the parameters themselves, with uncertainties
- Your data set is small or moderate

Machine learning

Reach for it when:

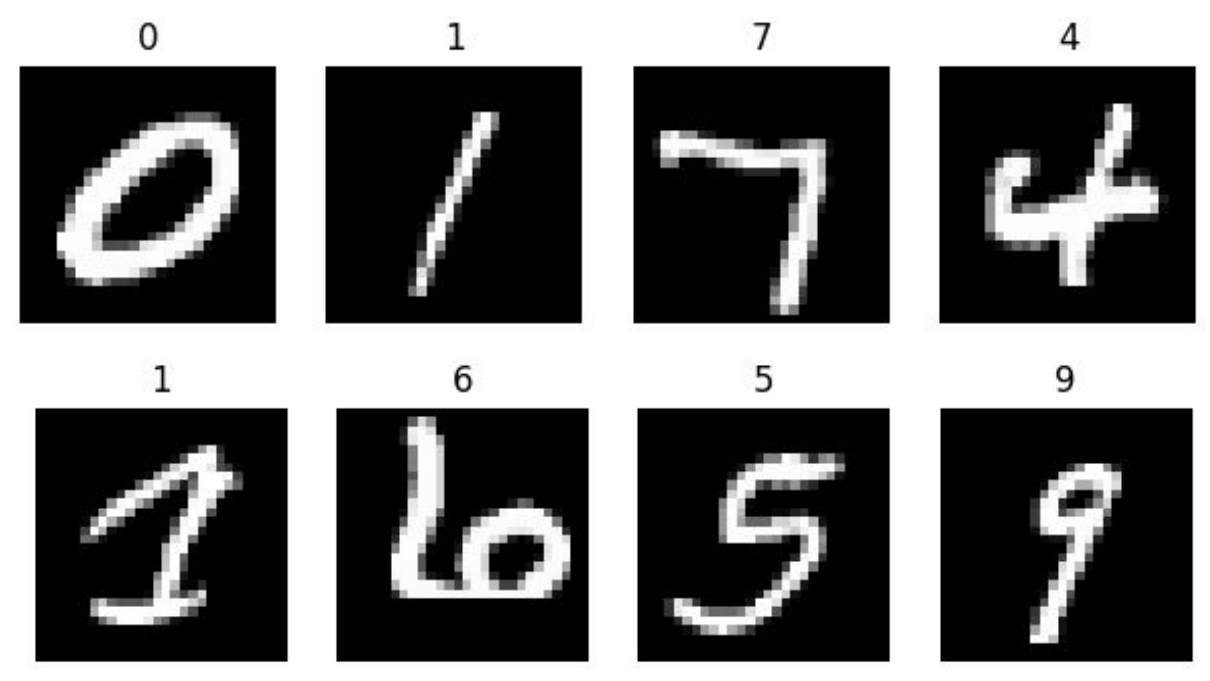
- There is no explicit model, or it is too slow to evaluate
- You want to sort, group, or predict across large data sets
- You have plenty of training data, real or simulated
- New data will resemble the training data

ML finds patterns and makes predictions within the range of what it has seen. It doesn't replace a physical model.

The ML Workflow: MNIST

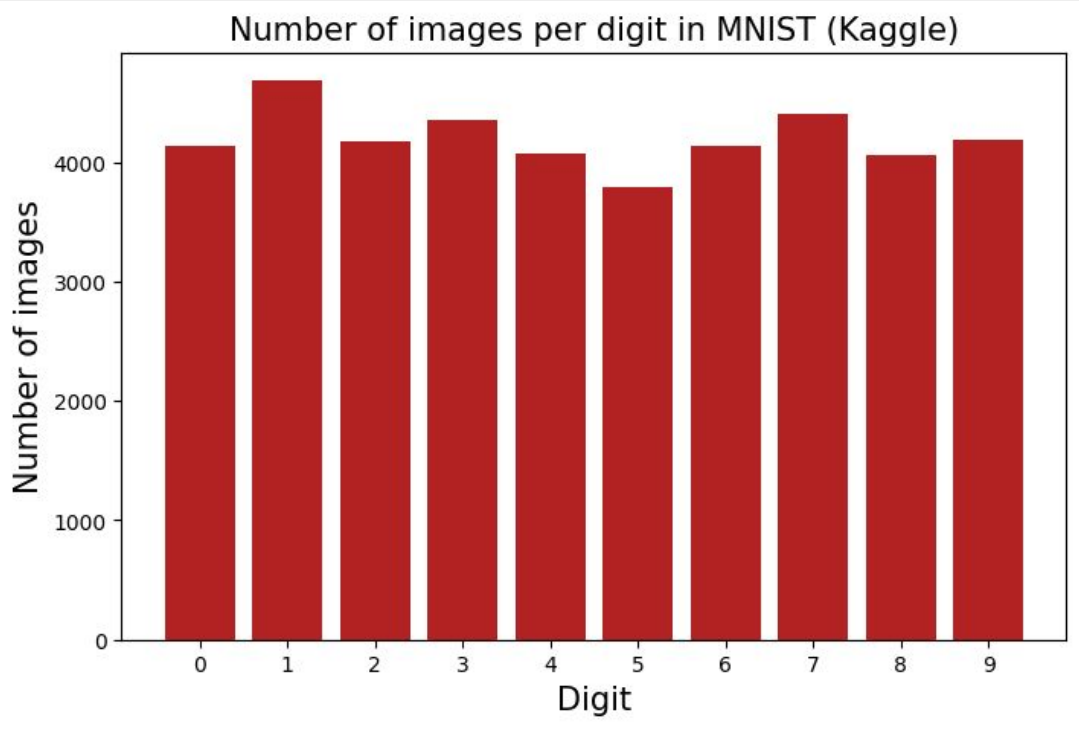
- What does your data look like?
- What task do you want to solve?

Inspect your data pt1



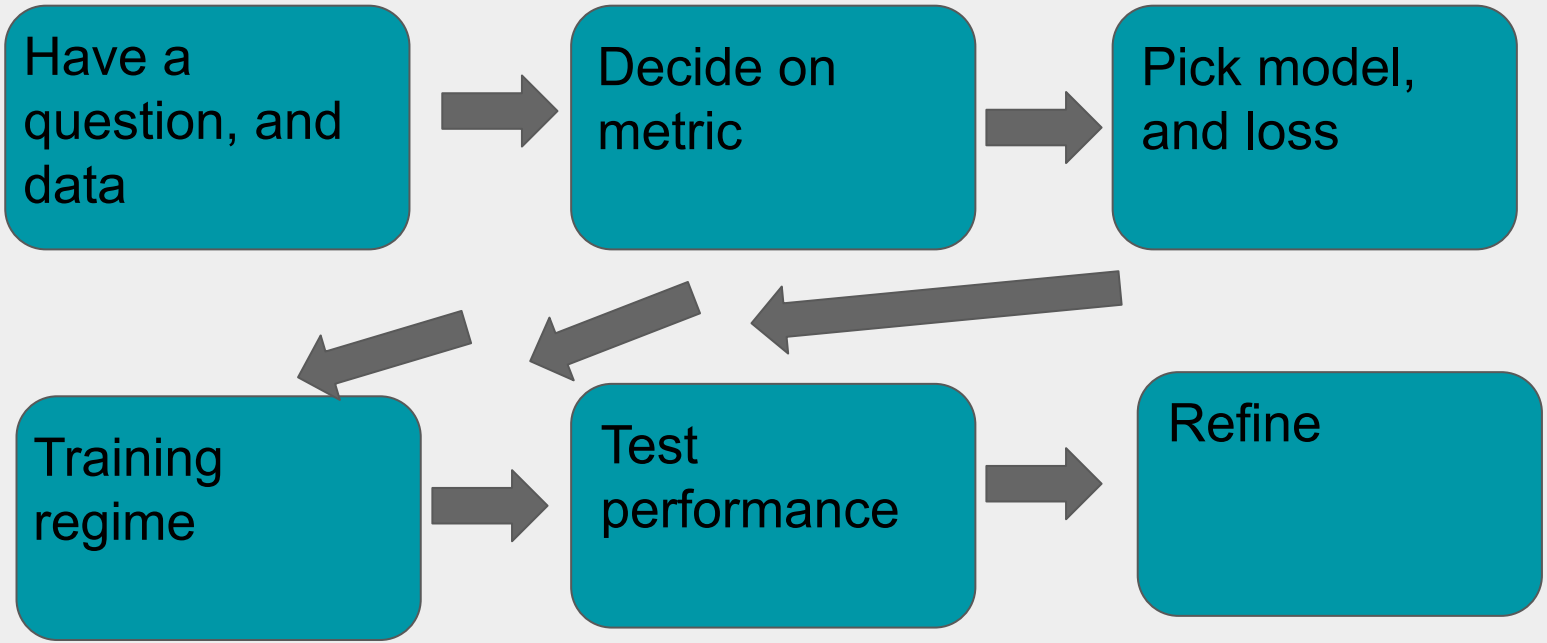
We want to classify hand drawn numbers

Inspect your data pt2



Check to see if any class is over or under represented

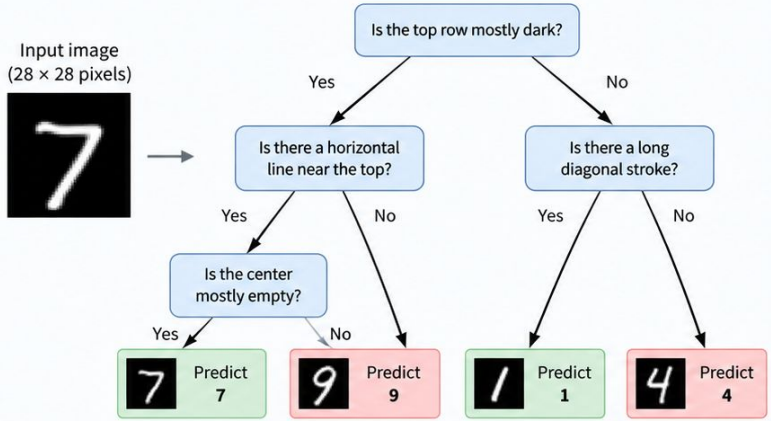
The ML Workflow (overview)



What type of model to use?

Decision Tree on MNIST

Classifies using hand-crafted pixel features and a series of if–else splits



Key Characteristics (Decision Tree)

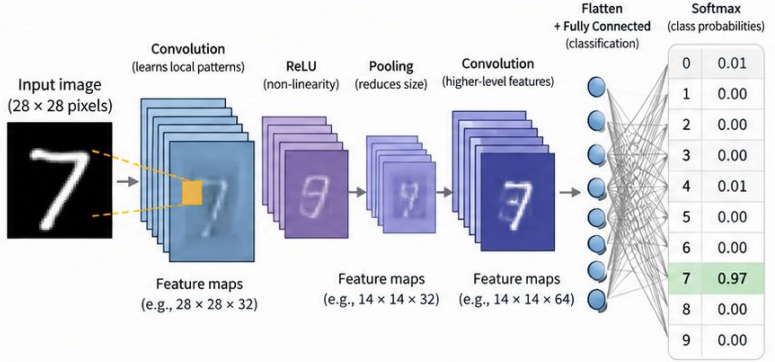
- Uses explicit rules on pixel values or simple image features (e.g., intensity sums, strokes, projections)
- Makes a sequence of if–else decisions (interpretable)
- No feature learning — relies on predefined features
- Can work on MNIST but typically lower accuracy than CNN
- Easy to visualize and understand

Example decision rule

```
If top_row_sum < 10
 and middle_row_sum > 50
 and bottom_row_sum < 30
  → predict 7
```

Convolutional Neural Network (CNN) on MNIST

Learns hierarchical features directly from pixel data



Key Characteristics (CNN)

- Learns hierarchical features directly from raw pixels
- Early layers detect edges and simple patterns
- Deeper layers combine patterns into digit-specific features (e.g., the shape of a 7)
- Typically much higher accuracy on MNIST
- Less interpretable, but very powerful for image data

Learned feature examples

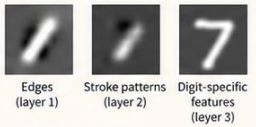


Image Credit: ChatGPT

Convolutional Neural Network (CNN)

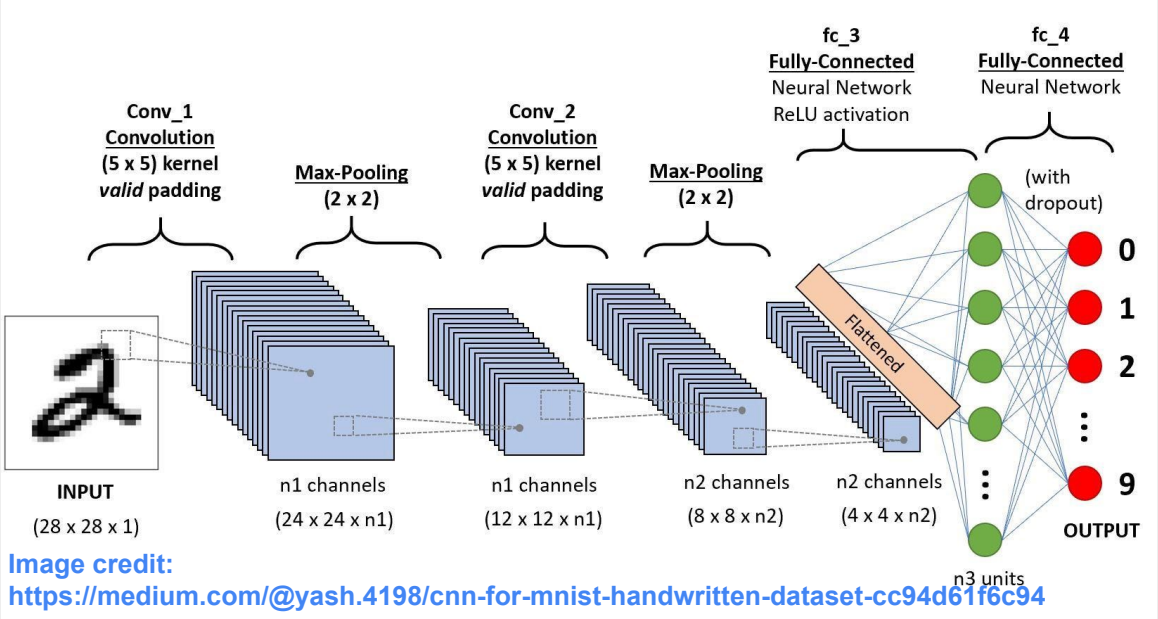


Image credit:
https://medium.com/@yash.4198/cnn-for-mnist-handwritten-dataset-cc94d61f6c94

Cross-entropy loss or mean square error?

Convolutional Neural Network (CNN)

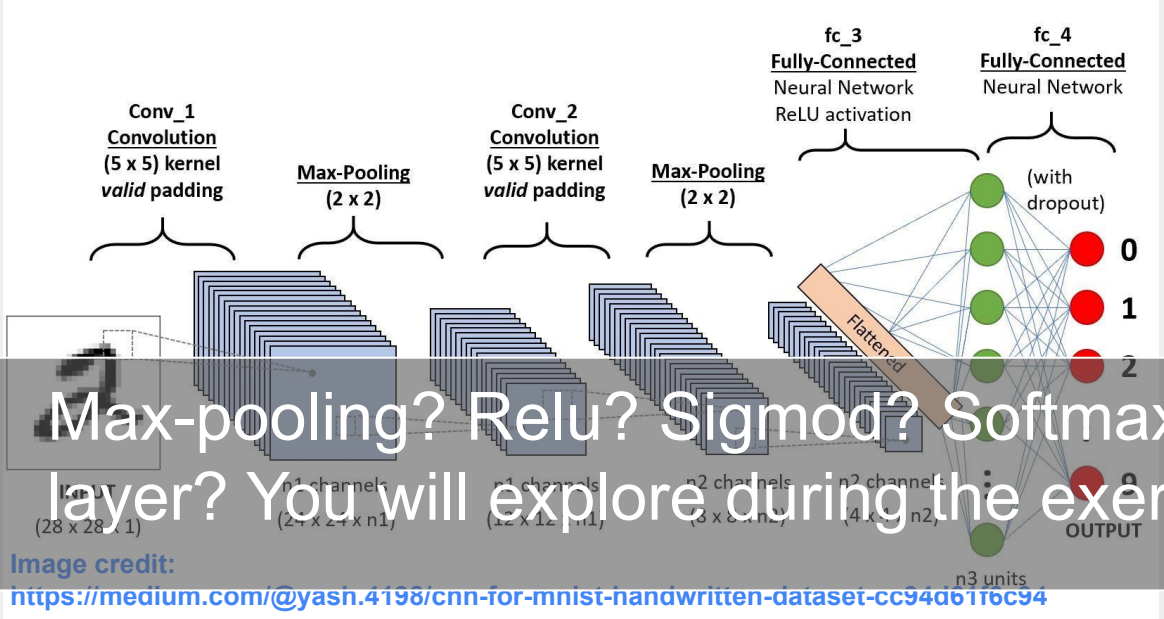


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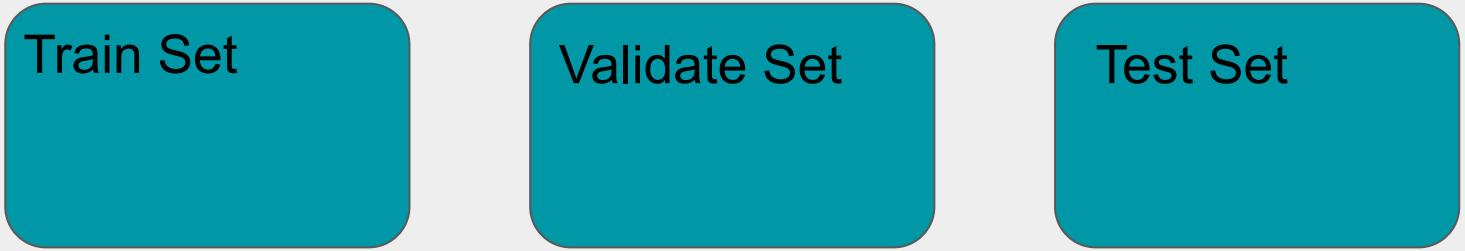
Cross-entropy loss or mean square error?

Max-pooling? Relu? Sigmod? Softmax Convolution layer? You will explore during the exercise!

Convolutional Neural Network (CNN)

- Graphic is scary, but its a multi-matrix model where the values of the matrices are “weights” (fitted params)
- Some extra steps in the start, and in-between each layer

Split your Data



- ~ 70% Train, 20% Validate, 10% Test
- Watch out for data leakage

The Loss Function AKA Cost Function



This is what we want to **minimize** to fit the model.

The Loss Function AKA Cost Function

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2$$

Cross-Entropy Loss (Multi-Class Classification)

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c})$$

Where:

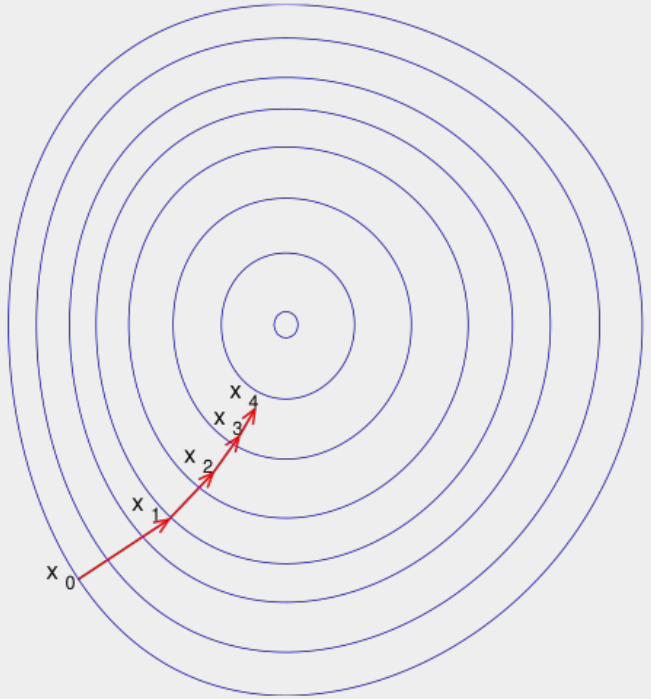
- N: Number of samples
- C: Number of classes (10 for MNIST)
- $y_{i,c}$: True label (1 if sample i belongs to class c, otherwise 0)
- $\hat{y}_{i,c}$: Predicted probability that sample i belongs to class c

This is what we want to **minimize** to fit the model.

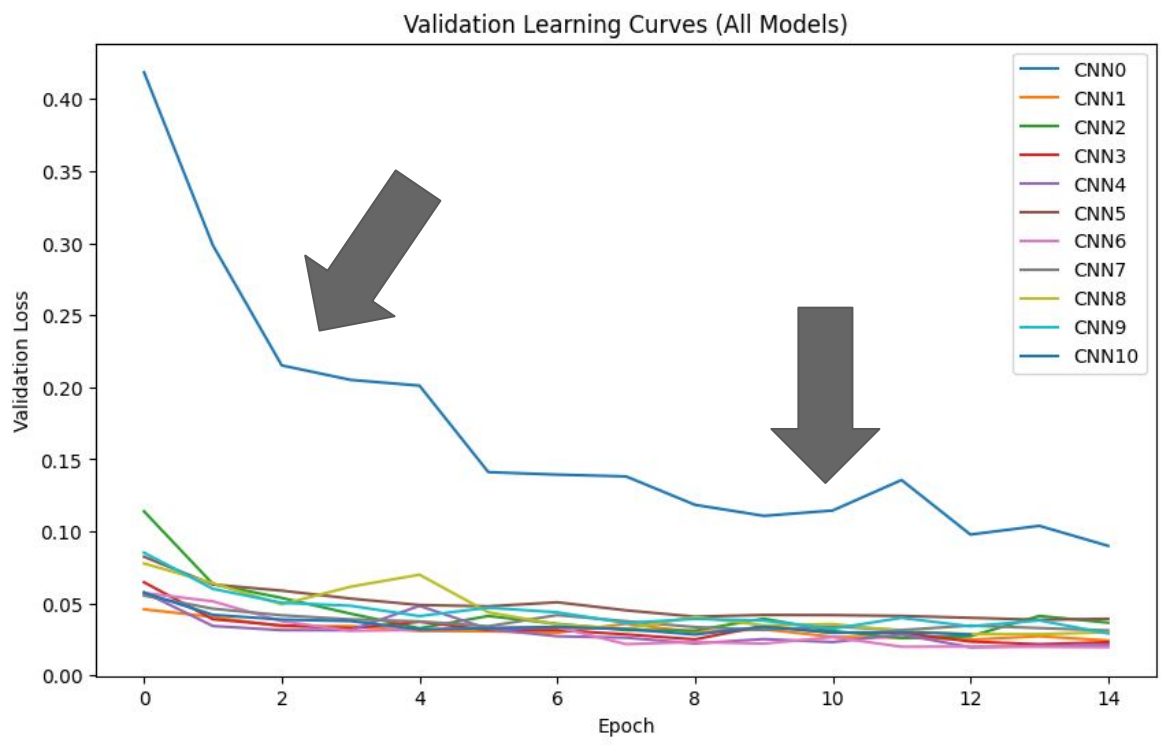
Gradient Descent

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_{\theta} L$$

- L is the loss function
- Theta are model params
- Eta is the learning rate
- Uses backpropagation & chain rule to calculate

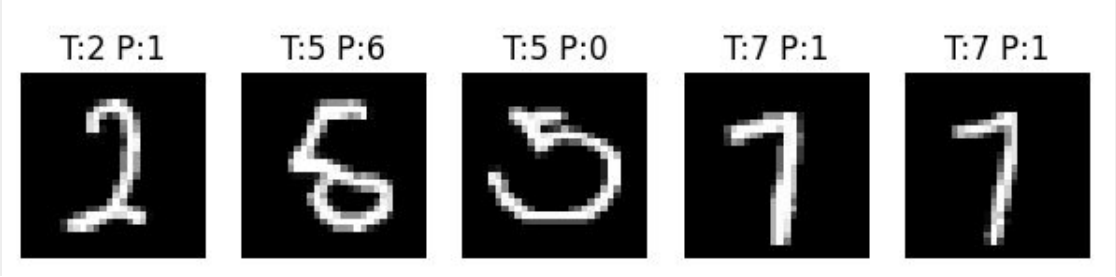


Understanding the training curve

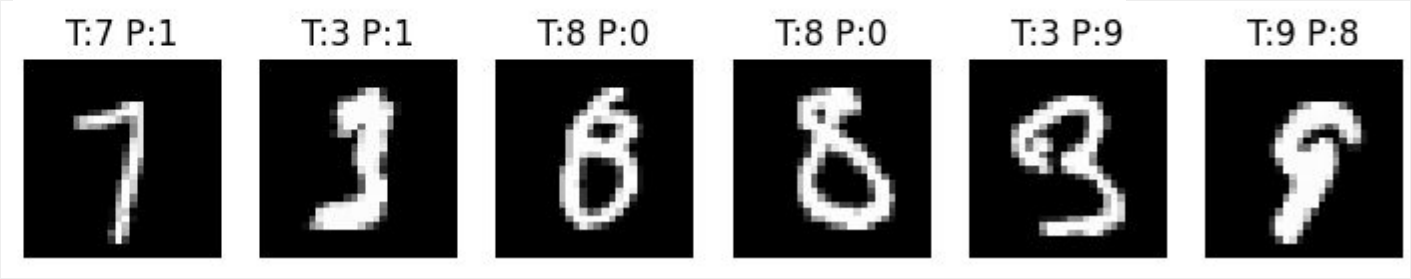


Initially steep, then flattens and oscillates

Overfitting Danger

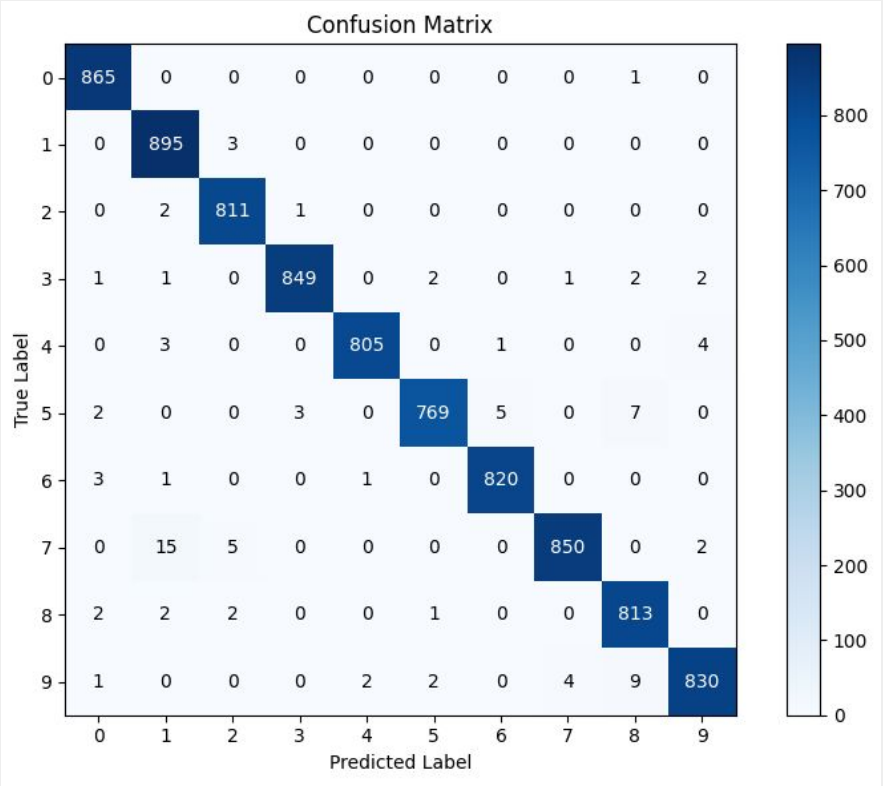


Harder to sort?
Or wrong labels?



Overfitting on the training data set is like memorizing the answer order for a test

Validation



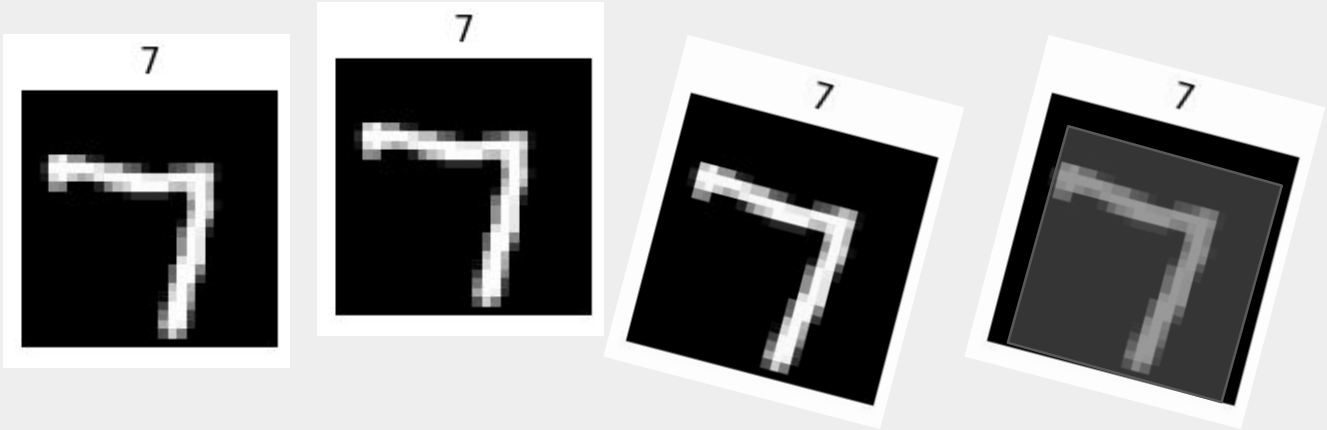
Overall correct % may obscure that the model is confusing some classes

Always split with training, and validation data

How can we improve?

- Use different class of model, choice of hyper-params
- Augment the training data
- Ensemble models

How can we improve?



We can make the CNN learn better if we add some offset, rotation, noise to the images => less memorization

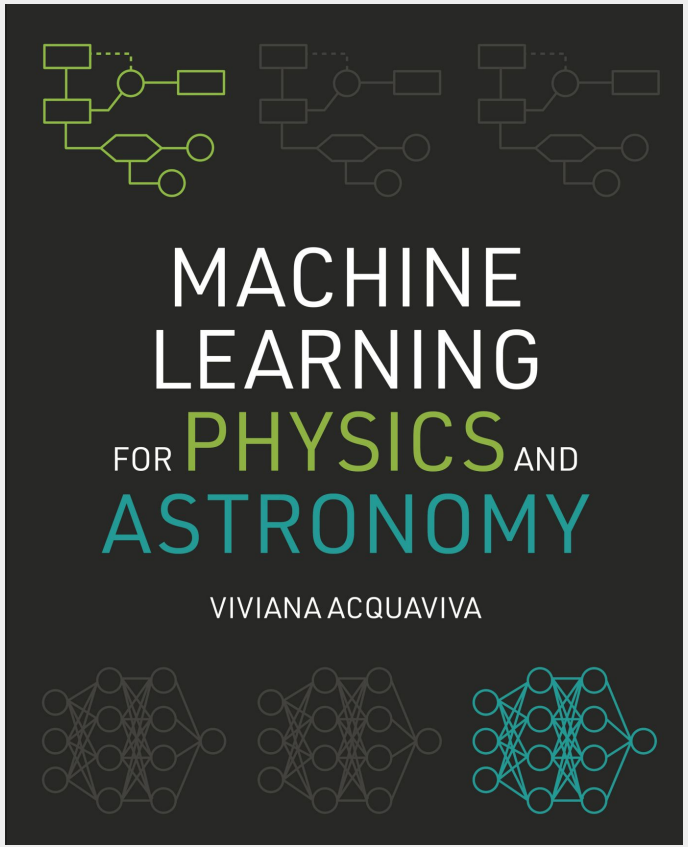
Not-NNs ML Models

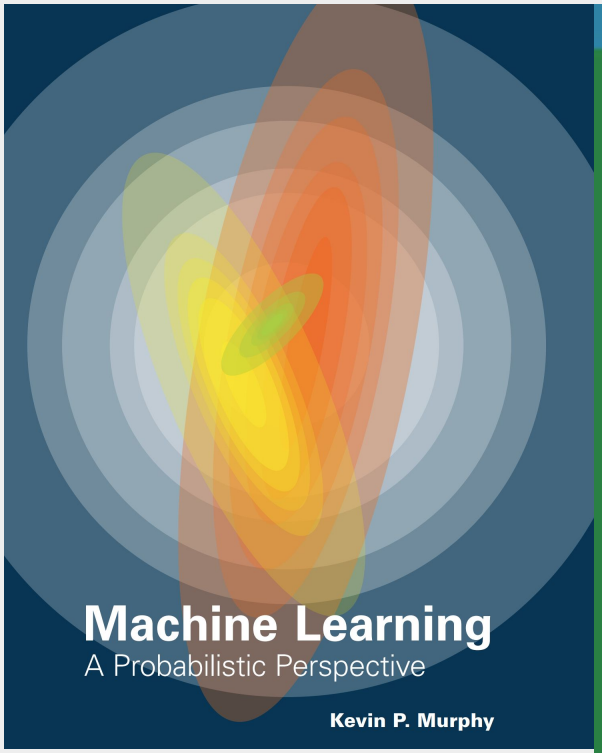
- Clustering
- Trees
- Gaussian Process Regression

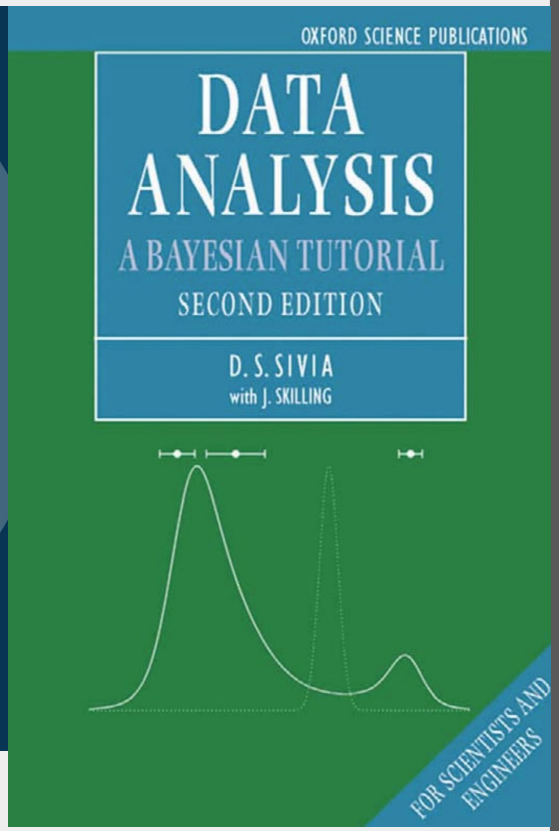
Similar enough:

- Transformers
- Principal component analysis

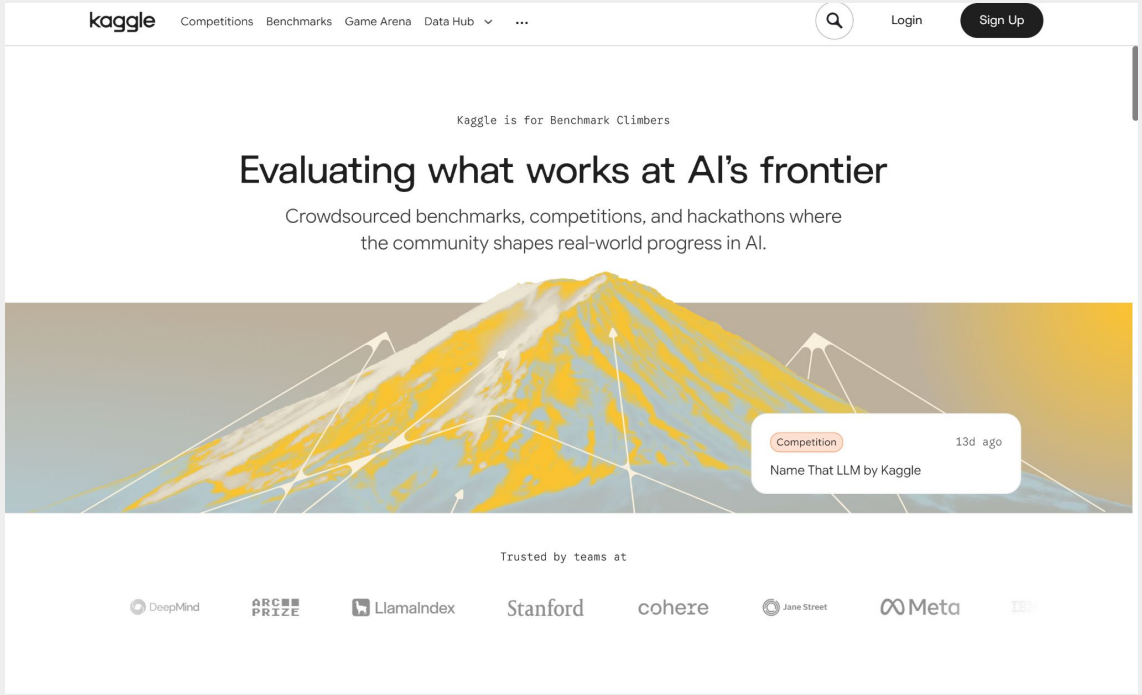
Recommended Textbooks







Kaggle is a ML competition website, winning teams post type ups of their methods



MNIST, and Galaxy Zoo are on there

In class activity!

- MNIST
- Best group gets timbits
- 🎄 vs 🧠
- How will you train? Hyperparameters?
- Presentations at the end



If it's too easy, we can download the full set from Kaggle